

Who belonged to the universe on that date?

Daily learning note | 2 October 2026 | Q009a

Sidiq review status: Not yet reviewed.

Research question and hypothesis

Can effective-date membership intervals recover a past universe, including securities that later exit? The hypothesis is that stable security identities and explicit entry/exit boundaries preserve historical membership without consulting a present-day roster. Q008 established split/dividend arithmetic; Q009a now establishes which identities belong at each date before features and targets. This is the next eligible non-Wednesday task. Terminal-event accounting remains separate in Q009b.

Beginner explanation and definitions

Imagine reconstructing a school's class list for January. Using only the students still enrolled in June would erase those who left. It might also add students who had not yet joined. Research universes have the same timing problem. Keep dated membership records instead of applying today's roster to every historical observation.

Universe: the set of eligible security identities under a declared rule. **Stable security ID:** a key intended to follow one security, independent of its display ticker. A company and a security are different levels of identity; one company can have multiple securities. **Effective date:** when a membership change applies under the source convention. **Entry:** first included date. **Exit:** first excluded date. **Re-entry:** a new membership interval after an earlier interval ends.

Coverage: the date range for which the caller asserts a complete history. **Survivor filter:** restricting a past population to identities present at a later date. **As-known-at history:** records limited to information available by a decision time. Effective dates alone do not establish this.

Data, metrics and evaluation protocol

All four identities and five intervals are synthetic. Coverage is January 1 through January 8, 2026; January 9 is the exclusive end. These are calendar-date labels, including non-session days, chosen to expose boundaries. They are not market observations or an exchange-session sample.

The evaluation uses eight explicit hand membership sets, an intentionally wrong final-date filter, invalid-input controls, re-entry and adjacency cases, an independent event ledger and deterministic subprocess replay. Membership comparisons are exact set/tuple comparisons with no numeric tolerance. Full project checks and a verified PDF are required. No empirical performance or alpha is estimated.

Mathematics and boundary convention

For security i with intervals indexed by j , define $M_i(d)=1$ if any interval satisfies $\text{entry}_{ij} \leq d < \text{exit}_{ij}$, and $M_i(d)=0$ otherwise. With no finite exit, membership continues only within declared coverage. The universe is $U(d)=\{i : M_i(d)=1\}$, evaluated only when $\text{start} \leq d < \text{end}$. Outside that range the answer is unknown and the API raises an error.

The half-open convention includes the entry date and excludes the exit date. It avoids counting two adjacent intervals twice: $[\text{Jan } 1, \text{Jan } 3)$ followed by $[\text{Jan } 3, \text{Jan } 5)$ includes Jan 3 exactly once. Same-ID overlaps are rejected rather than silently merged. A gap means nonmembership under the completeness assertion. Different security IDs may be eligible at the same time.

Worked synthetic history

Identity	Included from	First excluded date	Meaning
OLD	Jan 1	Jan 5	Later absent, still a historical member
STAY	Jan 1	No finite exit in coverage	Included throughout coverage
NEW	Jan 5	No finite exit in coverage	Enters when OLD exits
RETURN	Jan 2	Jan 4	First membership interval
RETURN	Jan 7	No finite exit in coverage	Re-enters after a gap

Date labels	Correct universe	Boundary illustrated
Jan 1	OLD, STAY	Initial entries included
Jan 2 and 3	OLD, RETURN, STAY	RETURN's first interval
Jan 4	OLD, STAY	RETURN's exit excluded
Jan 5 and 6	NEW, STAY	OLD exits; NEW enters
Jan 7 and 8	NEW, RETURN, STAY	RETURN re-enters

Deliberately wrong survivor control

At $d=\text{Jan } 2$, $U(d)=\{\text{OLD}, \text{RETURN}, \text{STAY}\}$. At $T=\text{Jan } 8$, $U(T)=\{\text{NEW}, \text{RETURN}, \text{STAY}\}$. The wrong filter $U(d) \cap U(T)$ returns $\{\text{RETURN}, \text{STAY}\}$: OLD is wrongly removed. One of three historical members is lost. Copying $U(T)$ directly would also include NEW before its entry.

No returns are attached to these identities. We therefore demonstrate the selection error without claiming a positive or negative performance bias. An exit can reflect many events; it is not itself a measured loss, liquidation instruction or zero recovery. A position ledger must retain holdings until supported accounting events resolve them.

Implementation and reproduction

src/quant_research/data/membership.py adds immutable MembershipInterval and MembershipHistory records. The history canonicalizes a tuple of records and members_on returns sorted immutable identity tuples. IDs reject blank values, whitespace and control characters. Entry, exit, coverage and query dates reject datetimes and strings; a datetime is a date subclass, so an exact date-type check is deliberate. No time or zone is discarded silently. Exit is required; None explicitly asserts continued membership.

Intervals must have positive duration and intersect coverage. They may extend outside its boundaries. Duplicate or overlapping intervals for one ID fail in either input order; adjacency and re-entry pass. An empty tuple explicitly declares an empty universe within coverage. The model cannot authenticate that declaration. Construction sorts in $O(n \log n)$; a query scans intervals and sorts selected IDs.

scripts/membership_experiment.py records the interval inputs, eight date sets and the wrong filter in experiments/membership/results.json. The experiment README documents the complete contract and limitations. Run make membership-experiment, make check and make note NOTE_DATE=2026-10-02.

Observed results and checks

Check	Observed result
Exact hand sets	All eight date sets match
Survivor-filter control	OLD wrongly removed from January 2 by final-date filtering
Independent event ledger	192 matches: eight dates across 24 input orderings
New Python suite	55 tests pass, including invalid input and boundary controls
Reproduction	Two subprocess runs match archived JSON bytes
Full project	470 Python tests and 66 C++ checks plus golden output pass
Code quality	Ruff lint/format passes on 41 files; strict mypy passes on 19 source files

The independent ledger removes exit events and then applies entry events on each date. It agrees with interval lookup through adjacency, gaps and re-entry. Separate tests check coverage failure, immutability, extreme dates, duplicate rejection and a future new identity leaving earlier results unchanged. All implementation checks passed on the first run. No coverage percentage was measured.

Primary sources consulted

CRSP Research Data Products, PERMNO and PERMCO, accessed October 2. The original CRSP page redirects to Morningstar. It motivates stable identity rather than display tickers; our synthetic IDs are not CRSP identifiers, and no vendor mapping or licensed sample is provided.
<https://indexes.morningstar.com/research-data-products/permno>

Python datetime documentation, accessed October 2. It explains date objects and the datetime/date subclass relation. Half-open membership and strict date-only inputs are our contract choices.
<https://docs.python.org/3/library/datetime.html#date-objects>

Limitations and interview explanation

This is a retrospective effective-date model. It lacks announcement times, revisions, processing latency and source vintages. A later correction to an earlier interval can still change past results. An interval history removes a current-roster error but does not prove point-in-time information safety. Q004a's licensed evidence audit and Q005b's provenance integration remain prerequisites for empirical use.

Coverage and stable identity are caller assertions. The API cannot detect a missing historical member, an incorrect ticker mapping or an unjustified open-ended interval. Missing exit evidence must not be converted to None unless continuation through coverage is established. Missing records are not proof of nonmembership. Completeness must come from audited upstream evidence.

Dates have no timezone or execution timestamp. Map exchange-local decision dates and source-specific before-open/after-close conventions upstream; this API does not validate sessions, tradeability, liquidity, price availability or fills. No market data, dependency, portfolio ledger or multi-universe framework was added. No performance or speedup claim is supported by this fixture.

For an interview, first distinguish effective membership, knowledge at the decision and actual holdings. Then state the interval boundaries, demonstrate OLD's exclusion by the wrong filter and explain unknown coverage. Finish by identifying what remains unproved: the history's completeness, publication vintage and terminal proceeds. Repository completion is separate from personal mastery or independent reproduction.

Next work and necessary input

Q009a is complete; Q009b terminal-event evidence is Ready for October 3. The Q009 parent remains incomplete. C003 remains the Wednesday October 7 C++ step. No input is needed for synthetic continuation. Real-data work still requires lawful entitlement and sample evidence. Existing access or a source/budget preference is needed before acquisition; do not send credentials. Personal review remains Not yet reviewed.

Daily runs remain at 08:00 Europe/London, including daylight-saving changes. Stop new research after 31 December 2026 and request the next programme scope. Publication authorisation is separate from review.

Check your understanding

1. Is a member with interval [Jan 2, Jan 4) included on Jan 4?
2. Why can a current constituent list corrupt a historical sample in two directions?
3. Does an effective-date interval prove the change was known at the research decision?
4. What should happen to a held security when it leaves the eligible universe?

Suggested answers

1. No. Entry is included and exit is the first excluded date; Jan 2 and Jan 3 are included.
2. It can remove securities that later exited and include later entrants before their entry dates.
3. No. Announcement times, revisions, latency and trustworthy source vintages are separate evidence.
4. Membership exit alone does not erase or value a holding. Preserve it until explicit transaction or terminal-event evidence supports an accounting change; unknown proceeds do not imply zero recovery.